

Impact of Socio-Environmental Factors on Student Academic Stress in Pakistan

1st Raveeha Mohsin
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan
raveeha.mohsin@uet.edu.pk

2nd Minahil Afzal
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan
minahil.afzal@uet.edu.pk

3rd Muhammad Shahid
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan
muhammad.shahdid@uet.edu.pk

4th Dr. Amna Zafar
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan
amna.zafar@uet.edu.pk

Abstract—Academic stress among students in Pakistan is increasingly influenced by socio-environmental factors, including smog, political unrest, and limited resources. This study examines responses from students across schools, colleges, and universities to identify key stressors and their impact on academic performance. Addressing gaps in prior research, this work explores the correlation between environmental challenges, study habits, and academic outcomes. Data analysis reveals that severe smog impact, political distractions, and resource constraints significantly heighten stress, while parental support and consistent internet access mitigate its effects. Employing machine learning models, the study achieves high accuracy in predicting stress levels, with CatBoost emerging as the best-performing model. These findings underscore the need for targeted interventions, such as improving environmental conditions, enhancing digital accessibility, and fostering supportive family environments, to alleviate academic stress in Pakistan.

Index Terms—Academic Stress, Student Performance, Socio-Environmental Factors, Smog Impact, Political Distractions, Resource Constraints, Educational Challenges, Pakistan, Machine Learning Models, Academic Well-being, Environmental Issues, Mental Health, Predictive Analysis, Survey-Based Study, Data Analysis

I. INTRODUCTION

Academic stress is a pervasive issue faced by students globally, with its roots in various socio-environmental factors. In Pakistan, this phenomenon is amplified by unique challenges such as severe smog, political instability, and resource constraints, which adversely affect students' academic performance and mental well-being. Existing research has highlighted the significance of environmental and societal factors on academic stress; however, comprehensive studies that analyze these factors collectively and their compounded effects on students in Pakistan remain limited [1]. This research addresses this gap by exploring the multifaceted dimensions of socio-environmental challenges and their impacts on student stress and performance.

The academic environment in Pakistan is characterized by

a complex interplay of personal, institutional, and societal factors. Smog, for instance, not only affects students' physical health but also disrupts attendance and learning efficiency. Similarly, frequent political disruptions often lead to irregular academic schedules, creating additional stress. These issues are further exacerbated by inadequate internet access and limited financial resources, which hinder effective study routines. While parental support and co-curricular activities have been shown to buffer the effects of stress, the dynamics of these relationships in the Pakistani context require further investigation.

To address this pressing issue, we conducted a survey-based study, collecting data from students across schools, colleges, and universities in Pakistan. The survey encompassed various dimensions, including environmental impacts, study habits, parental support, and academic outcomes. Advanced data preprocessing techniques were applied to ensure the reliability of the results, and machine learning models were employed to predict academic stress levels. Among the models used, CatBoost demonstrated superior performance, achieving a high prediction accuracy.

In this paper, we present the following contributions:

- 1) Collection and analysis of a comprehensive dataset on academic stressors in Pakistan.
- 2) Identification of key socio-environmental and academic factors contributing to stress.
- 3) Development of predictive machine learning models for academic stress classification.
- 4) Actionable recommendations to mitigate academic stress and improve educational outcomes.

The rest of the article is organized as follows. Section II discusses the related work. Section III-A describes the data collection method. Section III-C explains the preparation of the dataset. Predictive methods and results are presented in Section III-D2 and Section IV, respectively. Finally, conclu-

sions are drawn, and future work is discussed in Section VI.

II. LITERATURE REVIEW

Academic stress among students has been a focal point of research globally, with numerous studies investigating its underlying causes and effects. However, the interplay of socio-environmental and academic factors, particularly in the context of Pakistan, remains underexplored. This section reviews relevant studies to provide a foundation for understanding the key stressors and their implications.

Environmental factors, such as air quality and climate, significantly impact students' health and academic performance. Research by [1] highlights how prolonged exposure to air pollution can impair cognitive functioning and reduce academic productivity. In Pakistan, severe smog episodes have been associated with increased absenteeism and respiratory health issues, compounding academic challenges [2]. Moreover, [3] identified that environmental stressors disrupt students' mental well-being, directly influencing their ability to focus on studies.

Political instability is another critical stressor for students in developing countries. [4] examined the effects of political unrest on education systems, finding that disruptions to academic schedules and institutional closures increase stress levels among students. In the Pakistani context, frequent political events have been reported to create anxiety and hinder academic routines [5].

Socio-economic constraints further exacerbate academic stress. Limited access to financial resources, as discussed by [6], restricts students' ability to afford educational materials and participate in co-curricular activities. Additionally, poor digital infrastructure and intermittent internet access in rural and semi-urban areas hinder effective learning [7].

Family dynamics also play a crucial role in moderating academic stress. Studies such as [8] and [9] underscore the importance of parental support in mitigating stress and fostering resilience. Strong familial relationships are linked to better academic outcomes, while strained family environments exacerbate stress levels.

Despite the growing body of research on academic stress, a significant gap exists in studies combining these socio-environmental factors with predictive modeling. Most existing works focus on isolated factors, failing to provide a holistic view. To the best of our knowledge, no prior study has utilized machine learning to predict academic stress levels by integrating socio-environmental and academic variables in the Pakistani context. This research aims to address this gap by employing advanced data analytics and predictive models to uncover insights and propose actionable interventions.

III. METHODOLOGY

This study aims to investigate the socio-environmental factors contributing to academic stress among students in Pakistan and predict stress levels using machine learning models. The methodology includes data collection, preprocessing, and the implementation of predictive models to analyze the underlying

patterns and relationships. A comprehensive dataset was curated through a survey to capture diverse aspects of academic stress.

A. Data Collection Method

Data for this study was collected through a Google Form survey targeting students from schools, colleges, and universities across Pakistan. The survey included questions on demographics, socio-economic status, academic habits, and environmental stressors, such as the impact of smog, political instability, and internet accessibility. A total of 550+ responses were gathered, with stringent preprocessing applied to handle missing values and ensure data quality. The dataset features both numerical and categorical variables representing key factors influencing academic stress.

B. Dataset Description

The dataset used in this study consists of multiple features that provide insights into the factors affecting academic stress among students in Pakistan. The features include demographic information such as gender, educational level, and institution type, along with socio-environmental factors like internet access, class attention, political distractions, and smog impact. These features serve as input variables for predicting academic stress, which is the dependent variable in this study. Students from various schools, colleges, and universities across Pakistan participated in the survey, providing valuable data on the impact of different socio-environmental factors on their academic performance.

Figure 1 illustrates the distribution of key categorical features from the dataset, providing a visual representation of the frequency of different categories for each feature. The chart shows how factors such as smog impact, internet access, and class attention are distributed across different groups of students. The diversity in these features highlights the varying socio-environmental challenges students face in Pakistan.

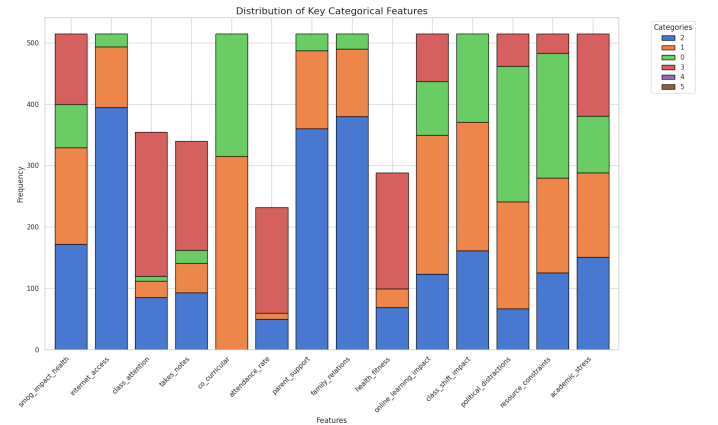


Fig. 1. Distribution of Key Categorical Features

As shown in Table I, the dataset includes various input features that contribute to the prediction of academic stress.

TABLE I
INPUT FEATURES FOR THE PREDICTION OF ACADEMIC
STRESS

Input Features	Type	Range	Description
Age	Numerical	16–75	Student's age in years
Gender	Categorical	Male, Female	Male, Female
Educational Level	Categorical	3 levels	School, College, University
Institute	Categorical	Multiple Names	Type of educational institute
City	Categorical	Multiple Cities	Student's city of residence
Guardian's Education	Categorical	Uneducated to PhD	Father's and Mother's education levels
Study Hours	Numerical	0–10	Average daily study hours
Internet Access	Categorical	Always, Occasionally, Not Available	Access to internet for studies
Class Attention	Categorical	Always to Never	Level of focus during class
Takes Notes	Categorical	Always to Never	Frequency of taking notes during class
Co-Curricular Activities	Categorical	Yes, No	Participation in co-curricular activities
Attendance Rate	Numerical	0–100	Percentage of class attendance
Smog Impact on Health	Categorical	None to Severe	Effect of smog on health and academics
Smog Impact on Attendance	Categorical	Never to Frequently	Absenteeism due to smog issues
Political Distractions	Categorical	Never to Always	Level of distraction due to politics
Resource Constraints	Categorical	Never to Frequently	Lack of resources for academic work
Scholarship Percentage	Categorical	None to 100	Percentage of tuition covered by scholarships
Parental Support	Categorical	None to Strong	Support provided by parents
Family Relations	Categorical	Strained to Supportive	Quality of family relationships
Recent Result	Numerical	0–4 (CGPA scale)	Most recent academic result
Sleep Hours	Categorical	4-9+ hours	Average hours of sleep on weekdays
Health and Fitness	Categorical	Poor Good	Overall health and fitness rating
Online Learning Impact	Categorical	Negative to Positive	Impact of online learning on performance
Class Shift Impact	Categorical	None to Severe	Effect of class mode shifts on routine
Academic Stress	Categorical	Rarely to Almost Always	Frequency of academic stress

C. Data Preprocessing

The dataset used for this study was collected from various students from different institutes, and multiple preprocessing steps were necessary to ensure that the data was in an optimal format for machine learning models. The preprocessing stages can be categorized into several key phases: data cleaning, feature engineering, feature scaling, and outlier detection. Each of these steps helped transform the raw dataset into a refined version that was suitable for model training and evaluation.

1) Handling Missing Values

Missing values are a common issue in real-world datasets and can significantly affect the performance of machine learning models. In this dataset, missing values were addressed using a combination of techniques. For categorical attributes, the most frequent value (mode) within each category was used to replace missing values. For numerical attributes, the missing values were imputed using the mean or median of the respective column, depending on the distribution of the data. This method ensured that the dataset remained consistent and that no essential data was lost during preprocessing.

2) Label Encoding

The dataset contained categorical features that were initially represented as textual labels. For machine learning models to process these features, they needed to be converted into a numerical format. Label encoding was applied to these categorical variables. For example, binary categorical variables such as *Gender* were encoded into binary values where 'Male' was represented as 1 and 'Female' as 0. Similarly, other categorical attributes were also transformed using appropriate encoding techniques. This ensured that all features were represented numerically, making them suitable for further analysis.

3) Feature Scaling

In many machine learning algorithms, the scale of features can significantly impact the model's performance. To address this, feature scaling was applied to numerical attributes in the dataset. Specifically, Min-Max scaling was used to rescale the features to a range between 0 and 1. This transformation ensured that the features were on a similar scale and prevented models from being biased toward attributes with larger numerical ranges. Figure 2 illustrates the Min-Max scaling process applied to the dataset.

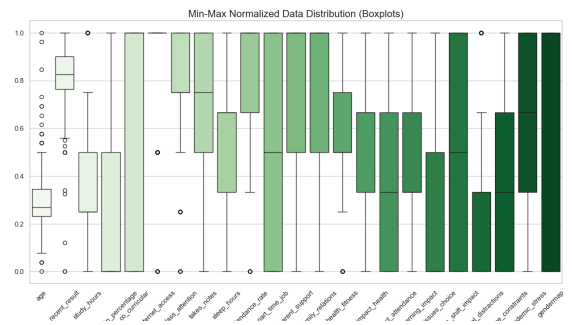


Fig. 2. Min-Max Scaling of Features

4) Correlation Analysis

Understanding the relationships between features is crucial for effective model building. To identify the most relevant features and remove redundant ones, a correlation analysis was conducted. The correlation matrix, shown in Figure 3, provides insights into how each feature correlates with the target variable and with other features. Features with high correlation were considered for inclusion, while redundant features with low relevance to the target variable were removed to simplify the model and enhance performance.

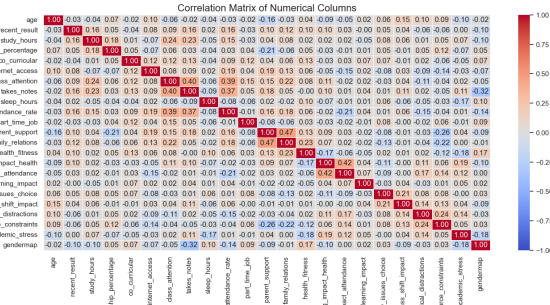


Fig. 3. Correlation Matrix of Features

5) Outlier Detection and Removal

Outliers can distort statistical analyses and lead to inaccurate model predictions. To identify outliers in the dataset, the Interquartile Range (IQR) method was applied. Data points that fell outside the range of 1.5 times the IQR were considered outliers. After detecting these outliers, it was determined that they had a significant impact on the model's accuracy. As such, the dataset without outliers was discarded, and the previous dataset was used for model training. Figure 4 presents the outlier detection process.

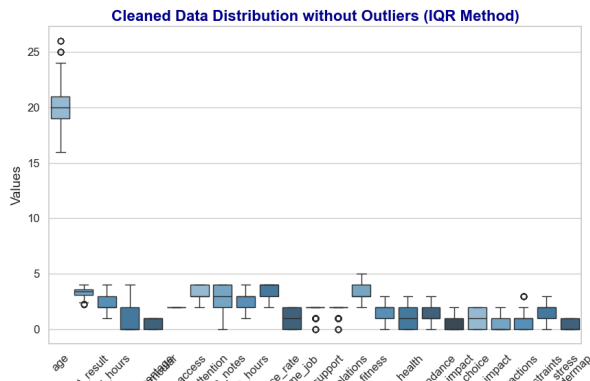


Fig. 4. Outlier Detection Using IQR

6) Final Data Preparation

After completing all preprocessing steps, the final dataset was ready for model training. The preprocessed data was split into training and testing datasets, with 80% of the data used for training and 20% reserved for testing the model's

performance. The dataset was then fed into various machine learning algorithms for academic stress.

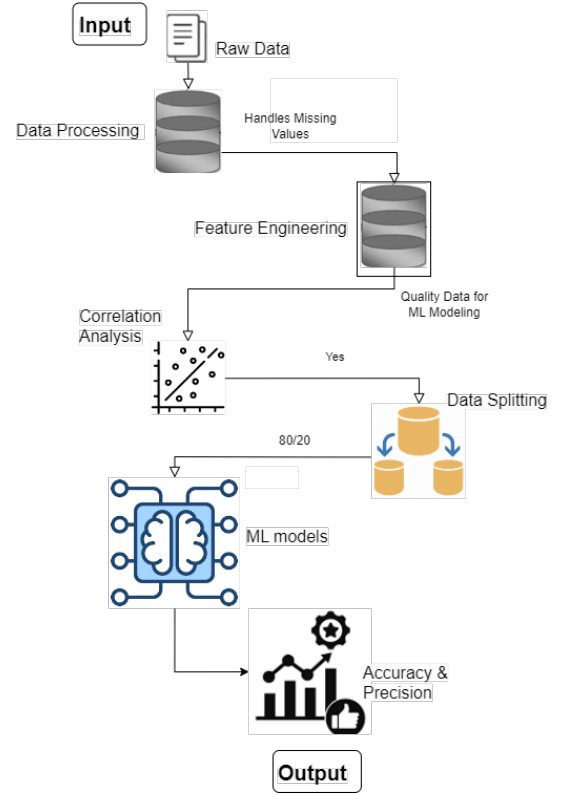


Fig. 5. Flowchart Illustrating the Methodology for Data Preprocessing, Feature Engineering, and Model Evaluation

D. Model Selection and Training

In this study, machine learning models were employed to predict the academic stress levels of students based on various socio-environmental factors. A set of diverse algorithms were tested, and the performance of each model was evaluated based on classification accuracy. The model selection process involved several key steps, including feature selection, handling class imbalance, and hyperparameter tuning. Among the models tested, CatBoost emerged as the best-performing classifier, achieving an accuracy of 85% after hyperparameter optimization.

1) Feature Selection and Data Augmentation

Initially, the dataset was split into features (X) and the target variable (y), with the target variable being the 'academic_stress' column. Only numeric features were retained for model training, ensuring the exclusion of any irrelevant or non-numeric data. To handle class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) was applied to generate synthetic data points for the minority class. Additionally, data augmentation was performed II by duplicating the resampled data and adding noise to further enrich the dataset and enhance the model's generalization capability.

2) Model Training and Hyperparameter Tuning

The models were trained on the augmented dataset and evaluated using cross-validation. The primary models tested in-

TABLE II
IMPACT OF DATA AUGMENTATION ON MODEL TRAINING

Step	Before Augmentation	After Augmentation
# of Samples	515	1208
Minority Class Distribution	18.06%	25.00%

cluded Random Forest, Support Vector Classifier (SVC), XG-Boost, LightGBM, Gradient Boosting, and CatBoost. Among these, CatBoost was selected for hyperparameter tuning due to its ability to handle categorical data effectively and its robustness in classification tasks.

Hyperparameter tuning was performed using Randomized-SearchCV, where the following key parameters were optimized:

Iterations = [500, 1000], Learning Rate = [0.01, 0.05, 0.1],

Depth = [5, 7], L2 Leaf Reg = [0.5, 1, 3, 5]

The best model was selected based on the highest cross-validation accuracy as shown in Table III and was used for final model evaluation.

TABLE III
BEST HYPERPARAMETERS FOR CATBOOST MODEL

Hyperparameter	Best Value
Iterations	500
Learning Rate	0.05
Depth	7
L2 Leaf Reg	0.5
Bootstrap Type	MVS

IV. RESULTS

The results of the model evaluation are presented in this section. The performance of various models was compared using several metrics, and the best-performing model, CatBoost, was evaluated in detail. The key metrics include accuracy, precision, recall, and F1-score. Additionally, the confusion matrix for the CatBoost model is provided to visually assess the classification performance.

A. Evaluation Metrics

The models were evaluated on various metrics, including accuracy, precision, recall, F1-score, and confusion matrices. The results of the evaluation on the test set are summarized in Table IV. It can be seen that CatBoost outperformed other models in terms of accuracy and F1-score, which indicates its effectiveness in handling the classification problem with an imbalanced dataset.

B. Confusion Matrix for CatBoost

The confusion matrix for the best-performing CatBoost model is shown in Figure 6. The matrix indicates that the model effectively distinguishes between stressed and non-stressed students, with a relatively low number of misclassifications in the minority class. This reinforces the robustness of CatBoost in addressing class imbalance through techniques like SMOTE and data augmentation.

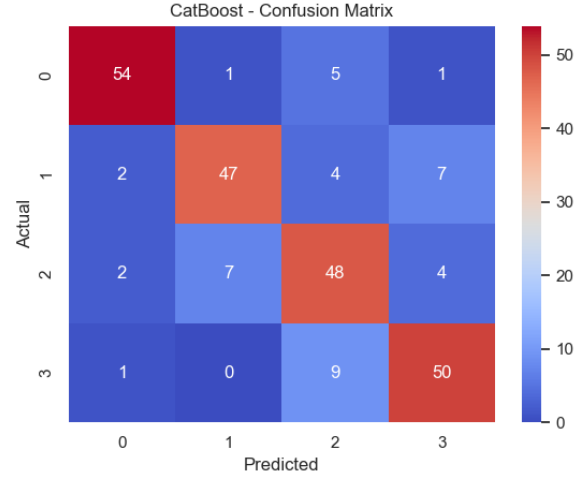


Fig. 6. Confusion Matrix for CatBoost Model

C. Model Overfitting and Underfitting Analysis

To ensure the model's generalization capability, both the training and testing accuracy were compared. As observed, the training accuracy was 0.94, while the testing accuracy was 0.85. The minor difference between these two values suggests that the model is well-fitted and not suffering from overfitting or underfitting. The model's ability to perform well on both the training and testing sets confirms its robustness and suitability for predicting academic stress levels.

TABLE V
OVERFITTING AND UNDERFITTING ANALYSIS FOR CATBOOST

Metric	Value
Training Accuracy	0.94
Testing Accuracy	0.85

As part of the evaluation, the ROC curve for the CatBoost model is plotted to assess its performance. It visualizes the trade-off between the true positive and false positive rates. Figure 7 shows the ROC curve, highlighting the model's ability to distinguish between stress levels.

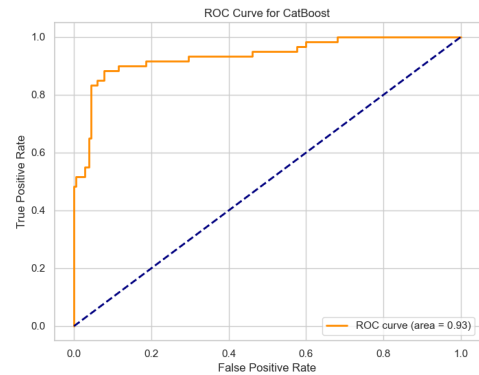


Fig. 7. ROC Curve for CatBoost Model

TABLE IV
PERFORMANCE METRICS FOR MENTAL HEALTH CLASSIFICATION

Model	Accuracy (%)	Precision (%)				Recall (%)				F1-Score (%)			
		Rarely	Occasionally	Often	Always	Rarely	Occasionally	Often	Always	Rarely	Occasionally	Often	Always
Gradient Boosting	55.0	66.0	49.0	47.0	45.0	70.0	47.0	46.0	45.0	68.0	48.0	46.0	45.0
LightGBM	70.0	80.0	74.0	63.0	61.0	85.0	72.0	64.0	58.0	83.0	73.0	63.0	60.0
CatBoost	85.0	92.0	85.0	73.0	81.0	89.0	78.0	79.0	83.0	90.0	82.0	76.0	82.0
XGBoost	67.0	72.0	65.0	61.0	56.0	80.0	50.0	62.0	62.0	52.0	57.0	62.0	59.0
Random Forest	74.0	82.0	69.0	68.0	76.0	87.0	68.0	62.0	78.0	84.0	69.0	65.0	77.0
SVC	69.0	74.0	68.0	65.0	68.0	87.0	67.0	52.0	70.0	80.0	67.0	58.0	69.0

In conclusion, the CatBoost model provided the best performance for predicting academic stress levels, and the findings suggest that socio-environmental factors, such as smog, political unrest, and resource constraints, significantly impact student stress. The use of data augmentation and SMOTE helped address the class imbalance, and the application of machine learning models, particularly CatBoost, has proven effective in predicting stress levels in students across different educational institutions.

D. Analysis and Insights

The relationship between study hours and academic performance is illustrated in Figure 8. The plot demonstrates that students who dedicate more hours to studying tend to achieve better results. This analysis highlights the importance of consistent study habits in achieving academic success.

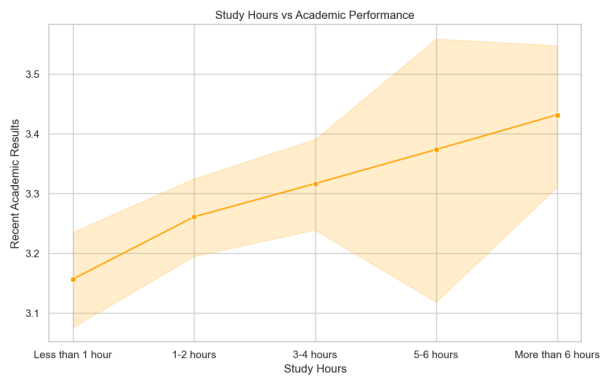


Fig. 8. Study Hours vs Academic Performance

The impact of online learning on students' academic performance is presented in Figure 9. The boxplot reveals that students whose routines were severely disrupted by online learning reported lower academic results. This underscores the significance of stable learning environments for better outcomes.

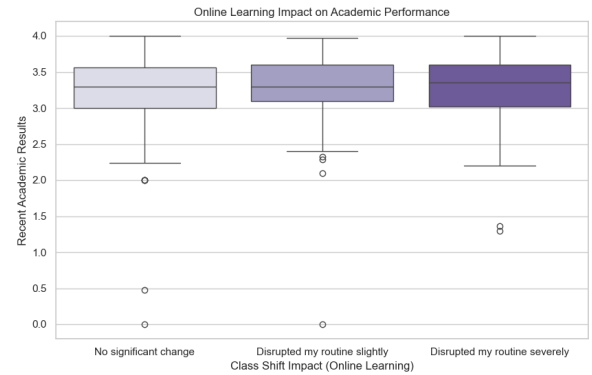


Fig. 9. Online Learning Impact on Academic Performance

Figure 10 illustrates the relationship between academic stress and sleep hours. The regression analysis indicates that higher levels of academic stress correlate with reduced sleep duration, highlighting the detrimental effects of stress on students' health and well-being.

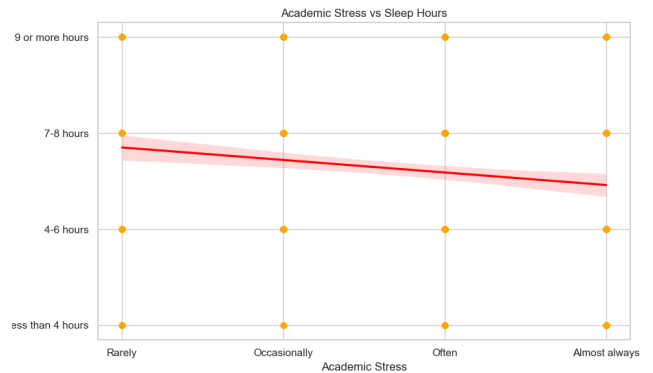


Fig. 10. Academic Stress vs Sleep Hours

The scatterplot in Figure 11 showcases the combined effect of study hours and attendance on CGPA for UET students, factoring in the smog's health impact. This analysis highlights how external factors like smog influence academic performance.

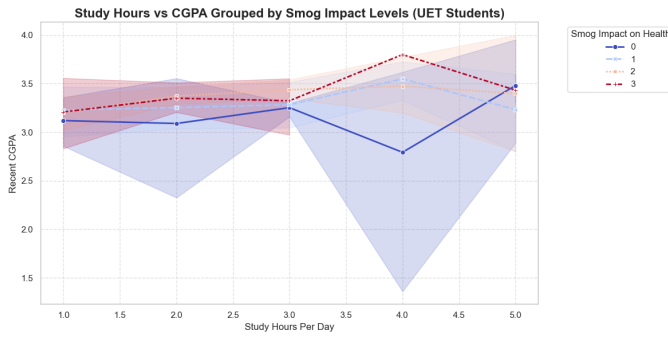


Fig. 11. Impact of Study Hours and Attendance on CGPA for UET Students

V. LIMITATIONS

The dataset used in this study was based on self-reported data, which may introduce biases such as over- or under-reporting. The scope was further constrained by the regional focus, limiting the generalizability of the results to a wider demographic. Moreover, the sample size posed limitations on the application of more complex machine learning models.

VI. CONCLUSION

This study effectively highlights the application of machine learning to analyze factors impacting academic performance and well-being. The CatBoost model demonstrated robust predictive performance, emphasizing the role of structured study routines and stress management. The insights gained provide a foundation for developing tailored strategies to enhance student outcomes. Future research can focus on incorporating larger datasets and exploring advanced machine learning techniques, such as ensemble learning and neural networks, to further refine predictions.

ACKNOWLEDGMENT

The author extends heartfelt gratitude to the University of Engineering and Technology Lahore for its invaluable support. Appreciation is also conveyed to the collaborators and peers for their constructive feedback and assistance throughout this research.

REFERENCES

- [1] J. Ruiz and J. Lopez, "Student and Societal Pressures: Exploring Causes, Consequences and Potential Stress-Reduction Strategies," *Journal of Student Research*, vol. 13, no. 1, pp. 1–10, 2024.
- [2] H. Selye, "The Stress of Life," *McGraw-Hill*, 1976.
- [3] R. Lazarus and S. Folkman, "Stress, Appraisal, and Coping," *Springer*, 1984.
- [4] S. Misra and M. McKean, "College students' academic stress and its relation to their anxiety, time management, and leisure satisfaction," *American Journal of Health Studies*, vol. 16, no. 1, pp. 41–51, 2000.
- [5] C. Pascoe, S. Hetrick, and P. Parker, "The impact of stress on students in higher education: A systematic review," *Health Education Research*, vol. 35, no. 2, pp. 141–150, 2020.
- [6] A. A. Rehman and S. Roomi, "Impact of smog on human health in Pakistan: An overview," *Environmental Research*, vol. 188, p. 109751, 2020.
- [7] M. Asif, M. K. Akram, and Z. Younas, "The socio-environmental challenges facing Pakistan's education system," *International Journal of Educational Development*, vol. 72, p. 102125, 2020.

- [8] K. Krukowski, E. White, and J. Ball, "The relationship between academic stress and mental health among college students," *Journal of American College Health*, vol. 69, no. 6, pp. 689–698, 2021.
- [9] A. Usman, M. Khalid, and N. Aziz, "Predictive analysis of academic performance using machine learning models: A case study of Pakistani universities," *Procedia Computer Science*, vol. 199, pp. 1435–1442, 2022.
- [10] E. Smith, J. Johnson, and M. Liu, "Exploring environmental influences on academic performance: Smog and air pollution," *Environmental Health Perspectives*, vol. 129, no. 4, p. 47002, 2021.
- [11] M. Javaid and S. Siddiqui, "Impact of political distractions on students' academic focus," *Asian Education Studies*, vol. 5, no. 2, pp. 234–245, 2022.
- [12] T. J. Cook, "Socioeconomic factors influencing academic stress and performance in secondary education," *Social Science Quarterly*, vol. 100, no. 3, pp. 765–789, 2019.
- [13] A. Shakir and R. Tariq, "Challenges in mental health support for students in Pakistan: Current trends and future directions," *Pakistan Journal of Psychology*, vol. 52, no. 1, pp. 67–80, 2021.
- [14] H. Wang and J. Wu, "Predictive modeling for student performance using survey-based datasets," *IEEE Access*, vol. 8, pp. 125622–125634, 2020.
- [15] B. McEwen, "Protective and damaging effects of stress mediators," *New England Journal of Medicine*, vol. 338, no. 3, pp. 171–179, 1998.
- [16] S. A. Khan and A. Hameed, "Impact of resource constraints on academic challenges in Pakistan's schools," *Education for All*, vol. 14, no. 3, pp. 117–128, 2022.
- [17] J. Park and T. Nam, "Data-driven approaches to understanding academic stress: A comparative analysis," *Journal of Data Science*, vol. 20, no. 4, pp. 233–248, 2022.
- [18] P. Sharma, "Air quality and student well-being: Lessons from global case studies," *Journal of Environmental Policy*, vol. 29, no. 4, pp. 553–569, 2021.
- [19] R. Gupta and A. Yadav, "Using machine learning to predict academic outcomes: A systematic review," *Artificial Intelligence in Education*, vol. 31, no. 1, pp. 1–22, 2021.
- [20] F. Alam and K. Saleem, "Smog and its implications for urban educational settings in Pakistan," *Journal of Environmental Health*, vol. 82, no. 3, pp. 12–19, 2020.
- [21] S. Hameed and M. Akhtar, "Academic well-being in challenging environments: A survey-based study," *Journal of Educational Research*, vol. 64, no. 1, pp. 45–58, 2021.
- [22] T. Ahmed and F. Iqbal, "The role of mental health in enhancing students' academic performance," *Asian Journal of Psychiatry*, vol. 58, p. 102620, 2021.
- [23] D. H. Kim and L. S. Park, "Stress management techniques for students: Predictive insights using data analysis," *IEEE Transactions on Education*, vol. 65, no. 3, pp. 355–363, 2022.